

Luyao Niu (she/her)

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EDUCATION

Peking University

M.Sc. in Urban Planning (Smart City and Big Data Program)

GPA: 3.75/4

Sept. 2023 – Jun. 2026 (expected)

Beijing Jiaotong University

B.Sc. Major in Traffic Engineering; Minor in Finance

GPA: 3.97/4 | Top 0.5% | Beijing Outstanding Graduate

Sept. 2019 – Jun. 2023

SELECTED RESEARCH EXPERIENCE

Joint Design of Shared Autonomous Vehicles and Public Transit Systems for Tourist Cities

- Formulated an integrated Shared Autonomous Vehicles (SAV)-transit design problem accommodating heterogeneous demand types (normal trips with carpooling and private tour trips requiring privacy), balancing system efficiency and accessibility in tourist city contexts.
- Applied continuum approximation and network queuing theory to derive flow equilibrium equations; formulated mixed integer program minimizing total system cost (agency and user costs) to determine optimal SAV fleet size and FRT network layout.
- Achieved 15-25% total cost reduction in low-demand scenarios compared to transit-only baselines; analyzed sensitivity of design outcomes to private tour trip proportions and demand levels; paper published in *Transportation Research Part C*.

Policy Impacts on Spatial Sentiment Sensing via LLMs and Interrupted Time Series Model

- Built computational framework leveraging LLMs to extract and quantify spatial identity dimensions (satisfaction, imagery, belonging) from 5+ years of cross-platform social media data in the Shenzhen-Hong Kong corridor.
- Integrated LLMs with Interrupted Time Series modeling to causally evaluate five regional integration policies, analyzing temporal dynamics and spatial heterogeneity across checkpoint, consumption, and socio-cultural areas.
- Quantified differential policy impacts: transportation infrastructure yielded immediate checkpoint satisfaction gains; development planning produced sustained effects in border spaces; identified overlooked implementation areas through spatial heterogeneity analysis; paper accepted by *EUGEO 2025 Oral* with another manuscript under review at *Applied Geography*.

Travel Behavior Shifts Following Electric Vehicle Adoption: A Big Data-driven Quasi-experiment

- Built large-scale quasi-experimental study using mobile signaling data to estimate causal impacts of EV adoption on travel behavior, overcoming limitations of small-sample survey methods in existing literature.
- Developed novel identification strategy leveraging APP usage logs (charging and brand-specific apps) to identify EV adopters as treatment group and construct matched control groups for Difference-in-Differences causal inference analysis.
- Revealed car usage rebound effects (strongest during off-peak and non-commute periods), increased active travel linked to charging infrastructure access, and metro substitution effects; co-authored manuscript under review at *Transportation Research Part D*.

Synthesizing the Impact of Public Transit Policy Interventions: A Meta-analysis

- Systematically reviewed 24 international studies on public transit policy interventions and identified inconsistent efficacy across different policy types, with limited understanding of underlying mechanisms.
- Executed meta-analytic modeling including effect size estimation, heterogeneity analysis, and moderator analysis to quantify intervention impacts and determine factors associated with policy effectiveness.

- Derived synthesized effect sizes for policy interventions including pricing strategies, service improvements, and information campaigns; identified implementation context and intervention intensity as key moderators of policy success; proceedings paper accepted by *AESOP 2025 Oral*; full journal version in preparation for *Transport Reviews*.

Optimized Path Planning and Traffic Flow Assignment Model Balancing User Equilibrium and System Optimum

- Developed a traffic assignment framework balancing user equilibrium and system optimum to mitigate congestion caused by homogeneous navigation recommendations; conducted SP survey and used discrete model to elicit heterogeneous preferences and constructed personalized alternative route sets for system-optimal assignment.
- Validated framework through simulation experiments on Beijing road network, demonstrating reductions in total travel time and congestion index; won **First Prize** in *National University Traffic Technology Competition*.
- Currently advancing research by applying inverse reinforcement learning on mobile signaling and trajectory data to better identify individual preferences and travel intentions; extending framework to connected and autonomous vehicle scenarios with centralized routing control.

Event-CausNet: Unlocking Causal Knowledge from Text with Large Language Models for Reliable Spatio-Temporal Forecasting

- Addressed limitations in spatio-temporal GNNs during non-recurring traffic events (accidents, work zones) by developing a causal inference framework that uses LLMs to extract and quantify causal knowledge from unstructured event reports, building a Causal Knowledge Base through Average Treatment Effects estimation.
- Engineered a dual-stream architecture integrating the Causal Knowledge Base with GNN-LSTM networks via novel Causal Attention mechanism to adjust traffic forecasts based on event-driven causal factors.
- Led full research execution from problem formulation to experimental validation; achieved 35.87% MAE reduction over state-of-the-art baselines on real-world datasets; paper under review at *IEEE IV 2026*.

Traffic Prediction via Signal Decomposition and Attention-based BiLSTM

- Addressed the challenge of disentangling trend, seasonal, and noise components in traffic flow data by developing MF-AttnBiLSTM, a hybrid framework combining moving average and discrete Fourier transform for multiscale signal decomposition.
- Designed a dual-stream BiLSTM architecture with multi-head self-attention to independently model temporal dynamics of decomposed trend and seasonal components, aggregating predictions for final traffic flow forecasts.
- Achieved state-of-the-art performance on benchmark PeMS04 and PeMS07 datasets; ablation studies quantified the contribution of signal decomposition stage to overall accuracy improvements; paper accepted for presentation by *IEEE GLOBECOM 2025*.

AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations

- Identified the Local Life Information Accessibility (LLIA) gap in 15-minute city initiatives, where existing location-based systems neglect spatial, temporal, and cognitive factors critical for hyperlocal decision-making and community engagement.
- Developed AskNearby, an LLM-powered community platform integrating a three-layer Retrieval-Augmented Generation (RAG) pipeline (combining graph-based, semantic-vector, and geographic retrieval) with a cognitive-map model encoding users' neighborhood familiarity and preferences for context-aware recommendations.
- Achieved significant improvements over LLM-based and map-based baselines in retrieval accuracy and recommendation quality through real-world deployments; demonstrated robust spatiotemporal grounding and cognitive-aware ranking capabilities; paper accepted for presentation by *ACM SIGSPATIAL GeoAI'25*.

Travel Route Choice Preference Analysis with Phone Signaling Big Data

- Engineered a machine learning framework (e.g., SVM, XGBoost) to analyze large-scale mobile signaling data and identify key factors influencing traveler route choice.
- Discovered significant route choice disparities under congestions and quantified the heterogeneity among different population groups.
- Independently executed the full research pipeline for the undergraduate thesis.

ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification

- Tackled the sparse-label (e.g., 5% regime) challenge in travel mode identification from GPS trajectories, addressing confirmation bias and manifold ignorance in prevailing semi-supervised learning methods.
- Developed ST-ProC framework, a multi-objective learning semi-supervised learning strategy integrating graph regularization, prototypical anchoring, and a margin-aware pseudo-labeling mechanism to actively reject noise and model intrinsic data structures.
- Achieved excellent performance on the GeoLife benchmark, demonstrated effectiveness of exploiting latent graph structure with margin-aware pseudo-labeling for trajectory classification, outperforming strong baselines by 21.5%; manuscript under review at *WWW Conference*.

Multi-Agent Scheduling and Collision Avoidance for Unmanned Warehouse Systems

- Formulated the multi-AGV task scheduling problem as a Colored Traveling Salesman Problem to maximize retrieval efficiency; implemented a hybrid solution utilizing the A* algorithm for pathfinding and a Mixed-Integer Linear Programming model for global optimization.
- Developed a dynamic zoning and load-balancing model using Monte Carlo Tree Search and UCT algorithms, optimizing the assignment of pallets to workstations to mitigate local congestion and minimize total travel distance.
- Engineered a collision avoidance strategy for dense environments using Deep Reinforcement Learning integrated with Convolutional Neural Networks for state feature extraction, effectively resolving multi-agent deadlocks; won **Second Prize** in *China MathorCup Modeling Competition*.

EXTRA-CURRICULAR ACTIVITIES

President, Student Union Beijing Jiaotong University	Jun. 2021 – Jun. 2023 <i>Beijing, China</i>
Volunteer, Beijing 2022 Winter Olympics National Indoor Stadium	Jan. 2022 – Apr. 2022 <i>Beijing, China</i>

SELECTED AWARDS & HONORS

National Scholarship (Highest undergraduate honor in China)	<i>Dec. 2020</i>
First-class Academic Scholarship (Top 1%)	<i>Dec. 2020</i>
First Prize, China University Traffic & Transportation Technology Competition	<i>Jul 2022</i>
Second Prize, China MathorCup Optimizing Mathematical Modeling Competition	<i>Apr. 2022</i>

SKILLS

Languages: Mandarin Chinese (Native); English (Fluent, TOEFL iBT 101); Spanish (Conversational)
Programming & Data: Python (Numpy, Pytorch, Scikit-learn, Geopandas, Networkx), C, MATLAB, R, SQL
Research Methods: Optimization, Machine Learning, Deep Learning, Reinforcement Learning, LLMs

PUBLICATIONS

Journal

[1] Shi, Z., Luo, S., **Niu, L.**, Shao, C. (2024). Joint Design of Public Transit and Autonomous Vehicle Mobility Systems for Tourist Cities Considering Abnormal Trips. *Transportation Research Part C*. doi: [10.1016/j.trc.2025.105179](https://doi.org/10.1016/j.trc.2025.105179).

- [2] **Niu, L.**, Wang, Y., He, S., Zhang, W.(2025). A Cross-Lingual Framework driven by Large Language Models for Spatial Identity Sensing from Social Media. (*Under review*)
- [3] Liu, C., **Niu, L.**, Wan, L., Zhang, W. (2025) Travel Behavior Shifts Following Electric Vehicle Adoption: Evidence from a Big Data-driven Quasi-experiment. (*Under review*)
- [4] Zhang, W., Wang, Y., Huang, N., **Niu, L.**, Wu, Y., Niu, C. (2025) Large-scale longitudinal mobility data reveal a causal view of social influence’s amplification after risk. (*Submitted to Nature Human Behavior*)

Conferences

- [1] **Niu, L.**, Wang, Z., Liu., J., Boukerche, A., Sun, P. (2025). MF-AttnBiLSTM: Traffic Flow Prediction via Hybrid Signal Decomposition and Dual-Stream Temporal Attention Learning. *IEEE GLOBECOM 2025, Oral Presentation*.
- [2] **Niu, L.**, Deng, Z., Li, B., Huang, N., Liu, R., Zhang, W.(2025). AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations. *ACM SIGSPATIAL GeoAI’25, Oral Presentation*. doi: [10.48550/arXiv.2512.02502](https://doi.org/10.48550/arXiv.2512.02502)
- [3] **Niu, L.**, Wang, Y., He, S., Zhang, W. (2025). Evaluating the Causal Impacts on Spatial Satisfaction Sentiment of Shenzhen–Hong Kong Regional Integration Policies: An Interrupted Time Series Study Leveraging Large Language Models. *EUGEO 2025 Proceedings, Oral Presentation*.
- [4] **Niu, L.**, Gu, J., Liu, C., Zhang, W. (2025). Why Public Transport Policies Work or Not? A Meta-analysis Based on the Behaviour Change Wheel. *AESOP 2025 Proceedings, Oral Presentation*. (*In preparation for submission to Transport Reviews*)
- [5] **Niu, L.**, Wang, Z., Guan, S., Sun, P. (2025) Event-CausNet: Unlocking Causal Knowledge from Text with LLMs for Reliable Spatio-Temporal Forecasting. (*Under Review*) doi: [10.48550/arXiv.2511.12769](https://doi.org/10.48550/arXiv.2511.12769)
- [6] **Niu, L.**, Huang, N.(2025) ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification. (*Under Review*) doi: [10.48550/arXiv.2511.13702](https://doi.org/10.48550/arXiv.2511.13702)
- [7] Zhu, H., Zhang, W., Huang, N., Li, B., **Niu, L.**,...,Liu, X. (2024). PlanGPT: Enhancing urban planning with tailored language model and efficient retrieval. (*Preprint*) doi: [10.48550/arXiv:2402.19273](https://doi.org/10.48550/arXiv:2402.19273)